

Modeling California Daily Maximum Temperature: Separating Seasonality and Multi-Decade Trends with Meteorological Predictors

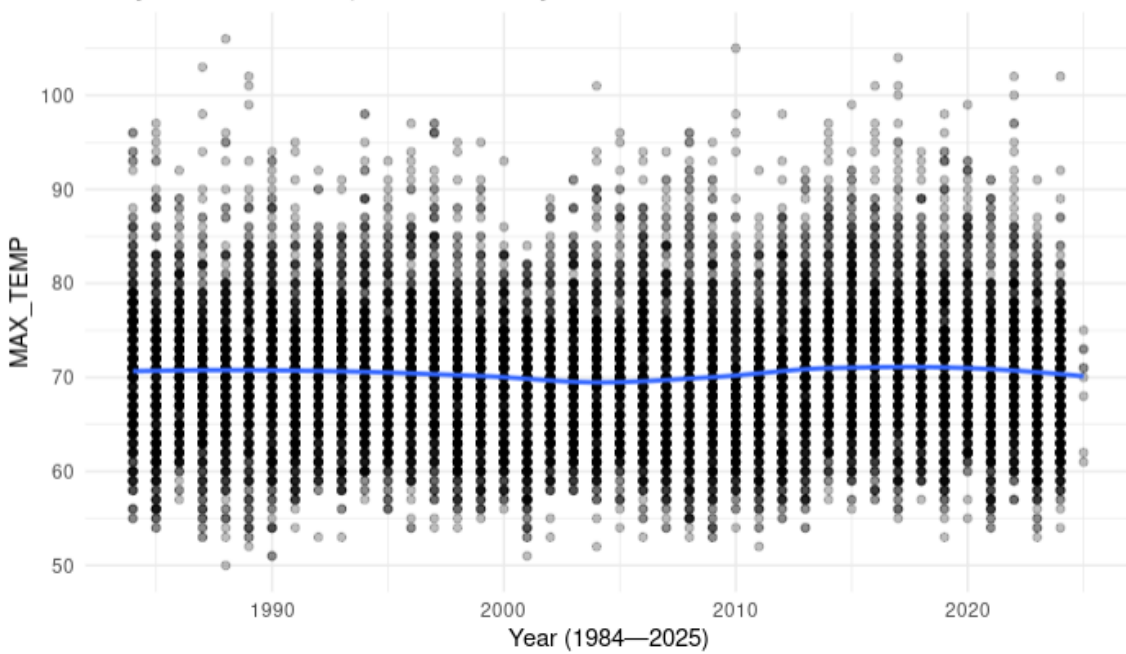
Motivation

California has experienced increasingly frequent and severe wildfires over the last several decades. Dry, hot seasons that are more prone to forest fires are lasting longer than before. Higher temperatures dry out vegetation and soils, and studies estimate that even a 1°C rise in temperature can increase the burned area by 600% in some western U.S. forests (NOAA, 2022). California’s extreme heat and wildfire risk are strongly tied to daily weather conditions. Understanding which meteorological factors drive daily maximum temperature, and how these effects vary across seasons and decades, can improve climate interpretation and support risk-aware planning. It leads to the question: how to predict future temperature trend based on meteorological conditions which affects seasonality change of maximum temperature, and its trend through last few decades.

Data Collection

We used the public dataset “California Weather and Fire Prediction Dataset (1984–2025) with Engineered Features” (Yavas et al., 2025), published on Zenodo. The dataset aggregates daily meteorological observations compiled from official sources (NOAA Climate Data Online) and aligns them with wildfire incident records (CAL FIRE), then provides engineered features designed for modeling.

Daily Maximum Temperature vs Day of Year



The multi-decade coverage allows us to estimate long-term temperature trends while separating them from short-term weather effects, the daily resolution captures strong seasonality in temperature, and the dataset compiles observations from official sources, it provides high-quality, standardized measurements with strong credibility. Overall, this dataset provides the right time scale, variables, and reliability needed to explain variation in daily maximum temperature and evaluate how meteorological effects relate to seasonal and long-term patterns.

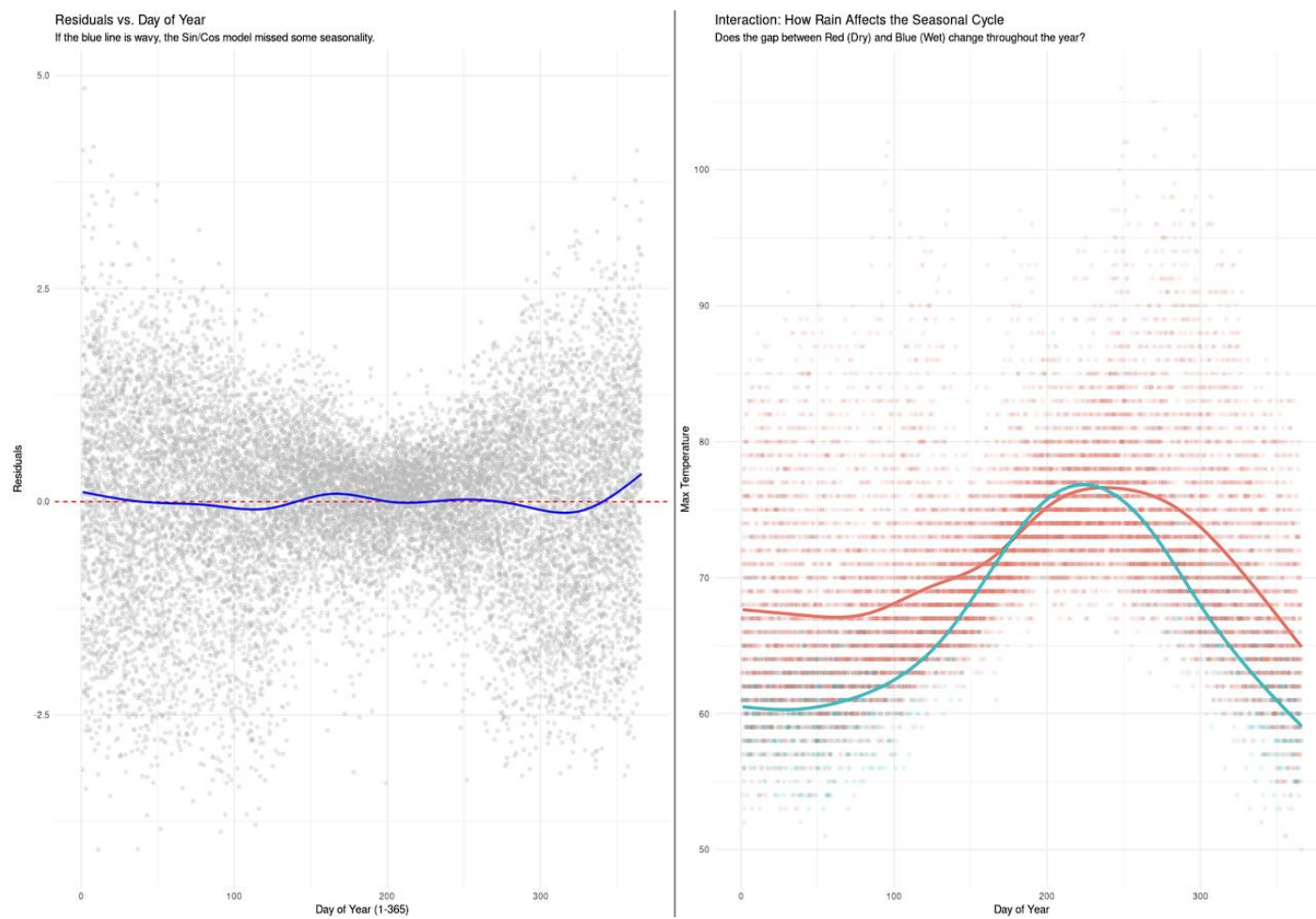
Methods of Analysis

Data processing

We removed records with missing values, coded FIRE_START_DAY as a categorical factor, and created YEAR_adj (years since 1984) to represent long-term trends. We also included average precipitation and average wind speed in a week as variables to increase accuracy of out prediction since it will also consider how short term trend will affect the near future.

Seasonality Capture Technique during the Empirical Data Analysis

Daily maximum temperature is strongly rely on the season, so we modeled day-of-year (DOY) as a periodical trend rather than a numerical value. From perspective of statistical model, it will only consider days as strictly ordered numbers, 365 is far from 1 and will result in a severely bias since there’s only one day between day 365and 1.We converted DOY to an angle $2\pi \cdot \text{DOY}/365$ and included Day_Sin = $\sin(2\pi \cdot \text{DOY}/365)$ and Day_Cos = $\cos(2\pi \cdot \text{DOY}/365)$. The peak and through in the image of Day_Sin + Day_Cos will capture the periodical trend of max temperature (peak at Aug and through at Jan) in years. Figure (Residuals vs DOY): the smoothed blue line stays close to 0 with minimal remaining “wave,” indicating that the Sin/Cos terms removed most seasonality so other predictors can be interpreted beyond the usual seasonal swing. Figure (Wet vs Dry seasonal curves): the wet and dry temperature curves have different slopes across the year, visually supporting season-dependent precipitation effects and motivating interaction terms.



```
# assumption validation methods
CData$MAX_TEMP_inv <- 1/(CData$MAX_TEMP^1.2)
CData$PRECIPITATION_trans <- 1/(CData$PRECIPITATION + 0.1)^3
CData$AVG_WIND_SPEED_trans <- log(CData$AVG_WIND_SPEED)
CData$LAGGED_PRECIPITATION_trans <- 1/(CData$LAGGED_PRECIPITATION + 0.1)^1.4
```

Transformations

To validate the slightly violated normality and constant variance assumptions, we applied a boxcox transformation to the response: MAX_TEMP_inv=1/(MAX_TEMP^1.2). We also transform other key predictors due to have non constant variance, like PRECIPITATION and AVG_WIND_SPEED, and an inverse-power transform for lagged precipitation with a small offset to handle zeros.

Results and Conclusions

Residuals:					
	Min	1Q	Median	3Q	Max
	-1.982e-03	-2.566e-04	4.345e-05	2.884e-04	2.358e-03
Coefficients:					
		Estimate	Std. Error	t value	Pr(> t)
(Intercept)		1.278e+02	1.666e+04	76.776	< 2e-16 ***
Day_Sin		5.103e-05	7.711e-06	6.618	3.77e-11 ***
Day_Cos		-8.681e-04	8.114e-05	-10.699	< 2e-16 ***
MIN_TEMP		-9.385e-05	1.985e-06	-46.846	< 2e-16 ***
AVG_WIND_SPEED_trans		-9.801e-05	5.307e-05	-1.847	0.0648 .
LAGGED_PRECIPITATION_trans		-5.912e-05	5.596e-06	-10.576	< 2e-16 ***
LAGGED_AVG_WIND_SPEED		-5.702e-05	4.585e-06	-12.436	< 2e-16 ***
FIRE_START_DAYTrue		-1.265e-04	1.017e-05	-12.433	< 2e-16 ***
PRECIPITATION_trans		-2.079e-06	1.202e-07	-17.296	< 2e-16 ***
YEAR_adj		3.003e-06	3.489e-07	8.605	< 2e-16 ***
Day_Cos:MIN_TEMP		2.755e-06	1.197e-06	2.301	0.0214 *
Day_Cos:AVG_WIND_SPEED_trans		2.717e-04	2.551e-05	10.650	< 2e-16 ***
AVG_WIND_SPEED_trans:PRECIPITATION_trans		6.726e-07	5.429e-08	12.389	< 2e-16 ***
LAGGED_PRECIPITATION_trans:PRECIPITATION_trans		8.985e-09	3.819e-09	2.351	0.0197 *
MIN_TEMP:LAGGED_PRECIPITATION_trans		8.271e-07	8.297e-08	9.969	< 2e-16 ***

Signif. codes:		0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1			
Residual standard error:		0.0004862	on 14961 degrees of freedom		
Multiple R-squared:		0.5772	Adjusted R-squared:		0.5768
F-statistic:		1459	on 14 and 14961 DF,		p-value: < 2.2e-16

Answer to Research Question: Interpretation of the coefficients and how predictors change the max temperature

Since our model transform the response reciprocally, the direction of the coefficients is inversed, for untransformed predictor, negative coefficients means positive influence.

- The Long-Term Trend (YEAR_adj) ----- As time progresses year over year.
Change in Max Temp: The coefficient is Positive, which means the baseline Maximum Temperature is slightly decreasing over the last few decades, after controlling for wind, rain, and fire events. This suggests that any observed warming in the raw data might be driven by changes in weather patterns (e.g., less rain) rather than a simple upward shift in the intercept.
- Daily Baseline (MIN_TEMP) ----- When the overnight low temperature is warmer.

Change in Max Temp: The coefficient is Negative. This means a warmer night leads to a significantly higher Maximum Temperature the following day. This is the strongest driver in your model.

Model specification & selection

We fit a multiple linear regression on MAX_TEMP_inv using meteorological predictors, seasonal trend controls (Day_Sin/Day_Cos), and scientifically motivated interaction terms. Candidate models were compared using stepwise procedures guided by AIC/BIC. AIC procedure gives exactly the same preliminary model we originally fitted, and BIC gives a smaller model without two interaction terms. To compare the goodness of two models, we then used ANOVA Overall Test and partial F-tests to verify whether removing terms (especially interaction blocks) caused a statistically meaningful loss of fit. When AIC and BIC disagreed by a small margin for both model(about 100 relative to a value around -22000), we selected the final model by prioritizing overall fit, that is we selected the model with the higher Adjusted R², indicating better explanatory power per parameter rather than overfitting by adding meaningless terms. And retaining meteorologically interpretable interaction structures, while enforcing the hierarchy principle.

Model validation

To guard against overfitting and assess generalization, we evaluated predictive performance using leave-one-out cross-validation (LOOCV). The final model showed stable LOOCV error consistent with its performance on the training dataset, supporting that our model predicts well to unseen days.

- Wind and Rainfall Conditions (AVG_WIND_SPEED_trans, PRECIPITATION_trans, and their lagged version) ----- For the days are dryer and has higher wind speed
Since the transformation of precipitation is inverse and for wind speed is ln, high value in precipitation means dryer days, high in wind speed means hotter days.
Change in Max Temp: The Negative coefficients for all of this conditions indicates that both higher wind speeds and lower precipitation leads to higher temperature. Which reflects to "Hot Wind" phenomenon and the cooling brake role rain played.
- Seasonality Condition (Day_Sin, Day_Cos, and corresponding interaction term) ----- Moving from Winter to Summer.
Change in Max Temp: The Sine/Cos terms create the expected annual wave (Hot Summer, Cold Winter).
In Summer(Cos = -1): The link between min temperature and max temperature is stronger. Wind drives heat even harder in the summer.
In Winter(Cos = 1): Wind has a weaker heating effect (or neutral) in winter. The link between is weaker.

Conclusion and Answer:

To predict future temperature trends in California, our analysis demonstrates that one must model the intensification of specific meteorological drivers rather than relying on a simple passive warming trend. Contrary to general expectations, the background climate baseline (YEAR_adj with possitive coefficients) reflects a slight cooling trend over the past 40 years once dynamic weather events are controlled for. Instead, the observed rise in extreme heat is driven by the specific conditions of aridity and wind dynamics.

Our model reveals that Maximum Temperature is most severely increased by Dry Conditions (lack of evaporative cooling) and High Wind Speeds (indicative of hot Santa Ana events). Crucially, these effects are seasonally dependent: wind-driven heating is significantly amplified during the Summer season, altering the shape of the seasonal temperature cycle. Furthermore, the compounding effects of warmer overnight lows (MIN_TEMP) and dry soil conditions create a feedback loop that exacerbates daily maximums. Consequently, future warming in this region will likely manifest through the increased frequency and intensity of these specific hot-wind and dry-spell events, rather than a uniform upward shift in the regional temperature baseline.

Limitation

- To improve normality and stabilize variance, we modeled a transformed response. Coefficients are therefore on the transformed scale, making effect directions less intuitive. In the future, we can report marginal effects back-transformed to the temperature scale to improve interpretability for non-technical audiences.
- Daily precipitation contains many zeros, so a single linear model may not fully capture the spike at 0 and the asymmetric effect of rainfall amounts. To refine this, we can use a two-part approach, logistic regression for rain/no-rain plus a continuous model for amount on rainy days, then combine predictions for total effect.
- We assumed stationary seasonality. Using fixed Day_Sin / Day_Cos assumes a stable seasonal cycle across decades, but climate change may shift the timing/amplitude of seasons. Improvement can be made by allow seasonality to vary over time and validate with time-aware cross-validation.

References:

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